P51

A matter of convention

Last class

one convention $\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$ — (1)

t - parameters

Our text book (Sec 2.4.2)

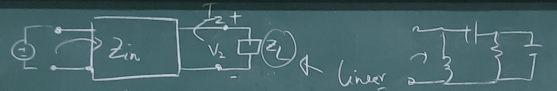
$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$ — (2)

ABCD parameters : on textbook

ABCD & t are 2x2 matrix

Last class what we did $t_{11} \equiv A$, $t_{12} = B$, $t_{21} = C$, $t_{22} = D$ (2) = { In other convention parameters in eq 1 are called ABCD

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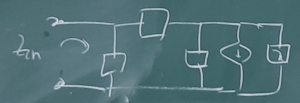


Linear

Z_L

Appr

Circuit equivalent



$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ (I_2 = 0 A), $Z_L = \infty$

$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

$-I_2 Z_L$ ($= I_1 \frac{Z_{21}}{Z_{22}+Z_L} = I_2$)

$V_1 = Z_{11} I_1 - \frac{Z_{12} Z_{21}}{Z_{22}+Z_L} I_1$

$\Rightarrow Z_{in} = \frac{V_1}{I_1} = Z_{11} - \frac{Z_{12} Z_{21}}{Z_{22}+Z_L}$

p53

$$\frac{V_1}{I_1}\Big|_{V_2=0} = \frac{1}{I_1/V_1}\Big|_{V_2=0} = \frac{1}{y_{11}}$$

$z_L = 0$

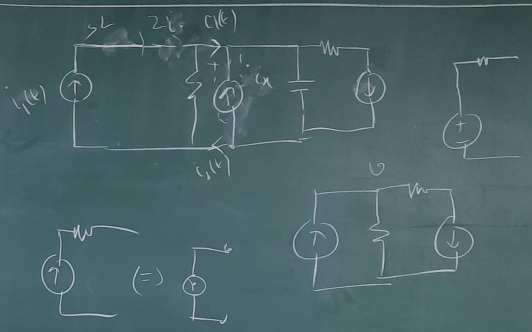
$$\left[\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}^{-1}\right]_{11} = \frac{a_{22}}{a_{11}a_{22} - a_{12}a_{21}}$$

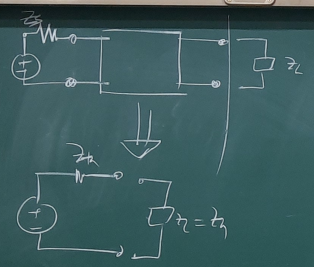
$z_L = 0 \Rightarrow z_{in} = z_{11} - \frac{z_{12}z_{21}}{z_{22}}$

$$= \frac{z_{11}z_{22} - z_{12}z_{21}}{z_{22}}$$

$$= \left(\left[[z]^{-1}\right]_{11}\right)^{-1}$$

$$= \left(y_{11}\right)^{-1}$$





$$-I_1 z_s = V_1 = z_{11}I_1 + z_{12}I_2$$

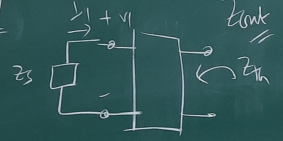
$$V_2 = z_{21}I_1 + z_{22}I_2$$

$$\frac{V_2}{I_2}$$

$$I_1 = -\frac{z_{12}I_2}{z_s + z_{11}}$$

$$V_2 = -\frac{z_{21}z_{12}I_2}{z_s + z_{11}} + z_{22}I_2$$

MPT



$$\Rightarrow z_{Th} = \frac{V_2}{I_2} = z_{22} - \frac{z_{12}z_{21}}{z_s + z_{11}} \overset{z_s=0}{=} \frac{1}{y_{22}}$$

(z_{out}) $\overset{z_s=\infty}{=} z_{22}$

p.4

$I_1 = Y_{11} V_1 + Y_{12} V_2$

$I_2 = Y_{21} V_1 + Y_{22} V_2$

$\downarrow$

$-V_2 Y_L$

$Y_{in} =$

Time-domain analysis



$$RC\frac{dV_c(t)}{dt} + V_c(t) = V$$

w/ $V_c(0) = 0$

$t \geq 0$

$\mathcal{L}$

$$RC\left(sV_c(s) - V_c(0)\right) + V_c(s) = V/s$$

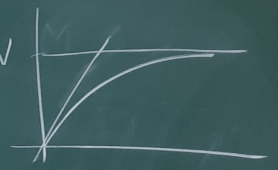
$$V_c(s) = \frac{V/s}{sRC+1} = V\frac{1/RC}{s\left(s+\frac{1}{RC}\right)}$$

$$V_c(s) = \frac{1}{RC} \Big/ s\left(s+\frac{1}{RC}\right) = \frac{1}{s} - \frac{1}{s+\frac{1}{RC}}$$

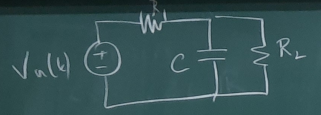
$$V_c(t) = \mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+\frac{1}{RC}}\right)$$

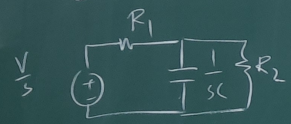
$\downarrow$

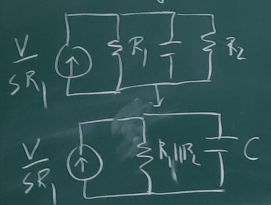
$$V_c(t) = V\left(1 - e^{-\frac{t}{RC}}\right), \quad t \geq 0$$



RC time constant









$$\Rightarrow 1 - e^{-\frac{t}{(R_1 \| R_2)C}}$$